Code	AMHM-ITAI-01
Title	Predicting Student Exam Performance
Version 1 Date	16 October 2024
Purpose	3rd Grade
Verification Date	12 December 2024

Assessment Material for Holistic Mission Code: EUHM−ITAI−01

Holistic Mission Title: Predicting Student Exam Performance

Mission Specifications:

You are an AI specialist at **Company X**, a leading educational technology company focused on improving student outcomes through data-driven insights. The company has recently partnered with several schools to explore and enhance student performance, and you've been assigned an exciting and impactful project: **Predicting Student Exam Performance**.

Your goal is to build a machine learning model that can predict student exam scores based on various factors. The dataset provided includes a range of features, including gender, parental education, and study time. The task will involve exploring and preparing the data, training and evaluating the model, and finally deploying it so that schools can use it to identify students who may need additional support.

This dataset includes information on students' scores in mathematics, reading, and writing, along with various socio-demographic features.

Features (columns):

- **Gender:** Male or Female
- **Parental Education Level:** Parent's highest education level (e.g., high school, college)
- **Lunch Type:** Standard or free/reduced
- **Test Preparation Course:** Whether the student completed a test preparation course
- **Study Time:** Hours of study per week
- **Absences:** Number of school absences
- **Math Score:** Score in the math exam
- **Reading Score:** Score in the reading exam
- **Writing Score:** Score in the writing exam

Elements of Holistic assessment
First Element: Explore and prepare the data
Evidence and proof requirements:

1. set Up the Environment
2. Inspect and clean the data
3. Prepare the data
4. Create visualizations to explore data relationships.
5. Use train_test_split to split the dataset into training and testing.

Performance Evidence		
#	Student's Performance (Steps)	Notes
1	Open Google Colab and login	
2	Load the dataset using pandas	
3	Print the first few rows to inspect the data	
4	Check for any missing values	
5	Handle any missing values (e.g., using fillna or dropna)	
6	Encode categorical variables (e.g., Gender, Lunch Type, Test Preparation)	
7	Normalize or scale numeric features like study time and absences	
8	Generate summary statistics of the dataset	
9	Create visualizations to show: The distribution of exam scores	
10	Create visualizations to show: Relationships between study time and exam scores.	
11	Create visualizations to show: how parental education impacts student performance.	
12	Use train_test_split from sklearn to split the data into 80% training and 20% testing	
13	Separate features (X) and labels (y).	

Second Element: Train and evaluate the model
Evidence and proof requirements:

1. Choose and load one of the models (Linear Regression, Decision Tree, Random Forest)
2. Fit the model on the training data (X_train and y_train).

3. Document the training process and potential challenges.
4. Generate Mean Squared Error (MSE) or Mean Absolute Error (MAE).
5. Plot predicted vs. actual values using a scatter plot.
6. Use GridSearchCV or RandomizedSearchCV to find optimal parameters for models.
7. Write a brief report on the same notebook

Performance Evidence		
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5	Plot predicted vs. actual values using a scatter plot.	
6	Use GridSearchCV or RandomizedSearchCV to find optimal parameters for models.	
7	Write a brief report on the same notebook about: Which model performed best and why.	
8	Write a brief report on the same notebook about: Key insights from the model (e.g., importance of features like study time, parental education).	

Third Element: Deploy the model

Evidence and proof requirements:

1. Install Gradio
2. Set up a Gradio interface that takes user input for features (e.g., gender, study time, parental education).
3. Consider that the interface should output the predicted exam score based on the trained model.
4. Run the Gradio interface locally and test it with sample inputs.
5. Ensure that the model predictions make sense based on user input by showing different screenshots from the results.

Performance Evidence

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2	Set up a Gradio interface that takes user input for features (e.g., gender, study time, parental education).	
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4	Run the Gradio interface locally and test it with sample inputs.	
5	Ensure that the model predictions make sense based on user input by showing different screenshots from the results.	

Assessment Methodology: performance evidence– product evidence
Required Resources: 1. **Internet Connection** 2. **Google Colab account** 3. **Dataset csv file**

Attached Files:

Required resources : Dataset CSV File Link :	student_performance_balanced missior
Assessment sheet	AS - EUHM-ITAI - 01.docx